

MD. MEHEDI HASAN

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RESEARCH INTERESTS

Behavioral machine learning, social engineering detection, interpretable AI, socio-economic modeling, continual learning, and geospatial risk prediction. Focused on building AI systems that are explainable, fair, and grounded in real-world human behavior.

EDUCATION

Dhaka International University

Sep 2023 – Dec 2027

Bachelor of Science in Computer Science and Engineering

CGPA: 3.76 / 4.00

BAF Shaheen College Jashore

Sep 2020 – Dec 2022

Higher Secondary Certificate

GPA: 4.75 / 5.00

PROJECTS

SocialAI – Autonomous Engagement Framework

- Engineered a distributed, scalable platform integrating LLM agents and resilient proxy routing for automated social media content generation and task execution at scale.

SafeRoads – Geospatial Risk Prediction

- Built a system identifying high-risk road segments using traffic, weather, and accident data.
- Supports urban planning and risk-aware navigation.
- Methods: Geospatial ML, XGBoost, Spatial Clustering*

MeteorShield – NASA Space Apps Challenge 2025

- Developed a planetary defense visualization platform using NASA NEO datasets.
- Champion (Barisal Division), Global Nominee, Honorable Mention**
- Focused on data storytelling and public awareness.

SaveFood – AI for Food Waste Reduction

- Trained time-series ML models to predict food spoilage from IoT sensor data (F1-score: 0.89).

AgroHub – IoT & ML Smart Agriculture

- Real-time farm monitoring and ML-driven decision support system for smallholder farmers to optimize yield.

DevBurn — Developer Analytics Engine

- A robust TypeScript-based full-stack engine designed to analyze developer workflows, integrating GitHub services with burnout tracking algorithms.

AI Learning Platform

- An AI-powered supplementary educational platform built to enhance university coursework with intelligent assistance and interactive tracking.

CryptoTracker

- A real-time cryptocurrency portfolio management application for tracking live market data and analyzing asset performance.

PUBLICATIONS

1. **Hasan, M. M., Rakib, R., Molla, M. A., Borhan, R., Based, M. A. A Socio-Economic Machine Learning Framework for Predicting Programmer Retention.** *Accepted at BIM 2025 (Taylor and Francis Books)* [\[Accepted\]](#)
2. **Hasan, M. M., Mahin, A. A., Chakraborty, S., Afrose, M., Mia, M. A., Based, M. A. Fraud Detection with DistilBERT A Transformer-Based Approach to Behavioral Banking Sequences.** *Accepted at BIM 2025 (Taylor and Francis Books)* [\[Accepted\]](#)
3. **Hasan, M. M., Mahin, A. A., Afrose, M., Based, M. A. MEWC-RL++: A Modular and Adaptive Continual Reinforcement Learning Framework for Automation Systems in Structured but Evolving Environments.** *Accepted at ECCT 2026 (Taylor and Francis Books)* [\[Accepted\]](#)
4. **Mrida, M. A. R., Hasan, M. M., Ausaf, S. M. N., Ray, P. C., Based, M. A. Impact of Social Media Usage on Academic Performance: A Hybrid Machine Learning and SEM Approach.** *Accepted at ECCT 2026 (Taylor and Francis Books)* [\[Accepted\]](#)
5. **Biswas, S., Hasan, M. M., Rakib, R., Mahin, A. A., Afrose, M., Based, M. A. Lightweight Deep Learning for Urban Traffic Intelligence: A MobileNetV2-YOLOv8 Pipeline with Adaptive Resolution Processing.** *Accepted at ECCT 2026 (ACSR, Atlantis Press, Springer Nature)* [\[Accepted\]](#)
6. **Asif, M., Hasan, M. M., Based, M. A. Federated and Explainable Machine Learning Framework for Cyber Attack Detection in Smart Grids.** *Accepted at ECCT 2026 (Taylor and Francis Books)* [\[Accepted\]](#)
7. **Hasan, M. M., Hossain, M. A., Mahin, A. A., Khatun, F. Student-YOLOv8: A YOLO-Based Deep Learning Approach for Real-Time Student Detection and Face Recognition in Classroom Attendance Monitoring Systems.** *Accepted at EFAST 2026* [\[Accepted\]](#)
8. **Hasan, M. M., Pallob, M. M. I., Based, M. A. Digital Transformation for Sustainable Universities: Modeling the Convergence of MIS, IoT, and Governance.** [Under Review (Journal)]
9. **Hasan, M. M., Mahin, A. A., Afrose, M., Based, M. A. War, Media Bias, and Socio-Economic Consequences: A Centennial Analysis of the Narrative-Reality Index (1914–2026).** [Under Review (Journal)]
10. **Hasan, M. M., Rahman, M. M., Mrida, M. A. R., Based, M. A. A Hybrid Explainable Machine Learning Framework for Customer Lifetime Value Prediction and Its Socio-Economic Implications in Emerging Digital Markets.** [Under Review]
11. **Rakib, R., Hasan, M. M., Molla, M. A., Based, M. A. A Relational Database-Driven Real-Time Cryptocurrency Portfolio Management System with Automated ETL Integration.** [Under Review]

12. Rahman, M. M., **Hasan, M. M.**, Based, M. A. **SafeRoads: AI-Driven Proactive Risk Synthesis and Data-Centric Infrastructure Audits for Road Safety in Low- and Middle-Income Countries.** [Under Review]
13. Molla, M. A., Rakib, R., **Hasan, M. M.**, Rion, A. M., Based, M. A. **Machine Learning based Regression and Classification of Earthquake Magnitude using USGS Seismic Records.** [Under Review]
14. Molla, M. A., **Hasan, M. M.**, Rakib, R., Borhan, R., Based, M. A. **Urban–Rural Inequality in Tech Career Intention: A Machine Learning Perspective.** [Under Review]
15. **Hasan, M. M.**, Molla, M. A., Rakib, R., Borhan, R., Based, M. A. **Socio-Economic Determinants of Programming Skill Development in Undergraduate CS Students.** [Under Review]
16. Rakib, R., Molla, M. A., **Hasan, M. M.**, Borhan, R., Based, M. A. **Explainable AI for Understanding Motivation in Choosing Computer Science Careers.** [Under Review]
17. Afrose, M., **Hasan, M. M.**, Mahin, A. A., Based, M. A. **Fairness-Aware HIV Risk Prediction in Bangladesh Using Behavioral and Explainable AI.** [Under Review]
18. Mahin, A. A., **Hasan, M. M.**, Rakib, R., Molla, M. A., Based, M. A. **A Modular Framework for Continual Reinforcement Learning in Dynamic Robotic Environments.** [Under Review]
19. Mahin, A. A., **Hasan, M. M.**, Afrose, M., Mia, M. A., Based, M. A. **Interpretable Machine Learning for Depression Risk Classification Among Bangladeshi University Students: Identifying High-Functioning Depression.** [Under Review]
20. Rion, A. M., Pallob, M. M. I., Rakib, R., Molla, M. A., **Hasan, M. M.** **Cross-Cultural Early Detection of Suicidal Ideation in Adolescents: An Ensemble Machine Learning Approach Using GSHS Data.** [Under Review]
21. **Hasan, M. M.**, Rakib, R., Molla, M. A., Based, M. A. **Women’s Psychology and Its Socio-economic Impact on Male Partners and Relationships.** [Ongoing Research]
22. **Hasan, M. M.**, Molla, M. A., Rakib, R., Based, M. A. **THE DUAL PARADOX: AI Investment and Workforce Displacement in Global Corporations.** [Ongoing Research]
23. Afrose, M., **Hasan, M. M.**, Based, M. A. **MEWC-LLM: Memory-Efficient Weight Consolidation for Large Language Models – Adaptive Stability–Plasticity Optimization for Factual Continual Learning.** [Ongoing Research]

ACHIEVEMENTS

NASA International Space Apps Challenge 2025 — Champion (Barisal), Global Nominee, Honorable Mention — [Team Polaris](#)
National Data Analytics Competition (NDAC) 2025 — Participant
Eastern Bank PLC Technovation’25 National Hackathon 2025 — Participant
BUET CSE FEST National Hackathon 2026 – Participant

PROFESSIONAL AFFILIATIONS

IEEE — Student Member (Valid through Dec 2026)
IEEE Computer Society — Student Member (Valid through Dec 2026)

BASIS – Student Member

LEADERSHIP & COMMUNITY

DIU Computer Programming Club - President	<i>April 2026 – Present</i>
DIU CSE Speakers Club — General Secretary	<i>May 2025 - March 2026</i>
DIU Career Development Club — Head, Research and Training Wing	<i>Aug 2025 – Dec 2025</i>
BASIS Students' Forum (DIU Chapter) — Executive Member	<i>Sep 2024–Oct 2025</i>

TECHNICAL SKILLS

Languages: Python, C, Java, SQL

ML Frameworks: Scikit-learn, TensorFlow, PyTorch, Auto-sklearn, LLMs / GenAI

Automation & Bots: Workflow automation, intelligent bots, web automation (Selenium, Playwright), API automation, task scheduling, RPA concepts, proxy engineering, logging & exception handling

Data Tools: Pandas, NumPy, Matplotlib, Seaborn

Dev Tools: Git, GitHub, VS Code, Google Colab, Kaggle

Specialized: Interpretable AI (SHAP), geospatial analysis, time-series forecasting, AutoML